PYTHON APPLICATION

TASK

1. Create a Jenkins pipeline to deploy this application on docker stack
2. GitHub URL: https://github.com/UmeshChandra2001/Python_Docker_App.git
3. Jenkins Stages
   - Code
   - Build
   - Code Quality Analysis (CQA)
   - Build Images
   - Scan Images
   - Push Images to docker hub
   - Deploy on Stack
4. create a cluster using swarm (1 manager 2 workers)
5. create a master-slave for Jenkins
6. run the pipeline on manager server

SOL:

Architecture:



Server & Service Details:

No.of Servers	Servers	Install services	Intance Type
1	Jenkins	jenkins,Sonar using Docker img	C7i
3 (DEV,UAT,PROD)	Manager	Git,Java 21 version,Docker,trivy	M7i
6 (DEV,UAT,PROD)	Worker	Docker	C7i

PULGINS NEED IN JENKINS:

- Pipeline: Stage View
- Docker Pipeline
- Sonarqube Scanner
- Quality Gates

SONARQUBE SETUP IN JENKINS

SonarQube Scanner plugin version 2.11 or later is required.

- From the Jenkins Dashboard, navigate to **Manage Jenkins** > **Manage Plugins** and install the **SonarQube Scanner** plugin.
- Back at the Jenkins Dashboard, navigate to **Credentials** > **System** from the left navigation.
- Click the **Global credentials (unrestricted)** link in the **System** table.
- Click **Add credentials** in the left navigation and add the following information:
    - **Kind**: Secret Text
    - **Scope**: Global
    - **Secret**: Generate a token at **User** > **My Account** > **Security** in SonarQube, and copy and paste it here.
- Click **OK**.
- From the Jenkins Dashboard, navigate to **Manage Jenkins > Configure System**.
- From the **SonarQube Servers** section, click **Add SonarQube**. Add the following information:
    - **Name**: Give a unique name to your SonarQube instance.
    - **Server URL**: Your SonarQube instance URL.
    - **Credentials**: Select the credentials created during step 4.
- Click **Save**

Add the webhook in sonarqube :

- Go to the project which you have created in SonarQube environment and webhook
- Navigate to the **Project Settings** > **Webhooks** > **create**
    - Give random **name**
    - **URL :** http://jenkinsurl:8080/sonarqube-webhook/
    - Click on Save

Add the webhook in Github :

- Go to the project which you have created in GitHub environment and Repo of the current project

- Navigate to the **Settings** > **Webhooks** > **Add webhook**
    - **Payload URL :** http://jenkinsurl:8080/github-webhook/
    - **Content type :** application/json
    - Click on **Add webhook**

Docker Part:

- Please provide permission to the all manager server (chmod 777 /var/run/docker.sock)
- Add Docker hub Credentials in Jenkins
    - Go to **Manage Jenkins** > **Credentials**
    - Click **Add credentials** in the left navigation and add the following information:
        - **Kind**: Username and Password
        - **Username:** Provide username of docker hub
        - **Password:** Provide password of docker hub
        - **ID & Description:** Give the credentials under what id u want to save
- Add docker hub credentials in pipeline in with ID

Install trivy for manger server :

CMDs:

Step 1: Add the Official Trivy Repository

```
cat << EOF | sudo tee /etc/yum.repos.d/trivy.repo
[trivy]
name=Trivy repository
baseurl=https://aquasecurity.github.io/trivy-repo/rpm/releases/\$basearch/
gpgcheck=1
enabled=1
gpgkey=https://aquasecurity.github.io/trivy-repo/rpm/public.key
EOF
```

Step 2: Install Trivy

```
sudo yum -y install trivy
```

Step 3: Verify the Installation

```
trivy –version
```

Jenkins Pipeline automation:
DEV pipeline > UAT pipeline > PROD pipeline

Created 3 Types of NODEs:

- LMS_DEV
- LMS_UAT
- LMS_PROD

Pipeline Scripts are pushed to GIT Repo:
Automated pipeline using:

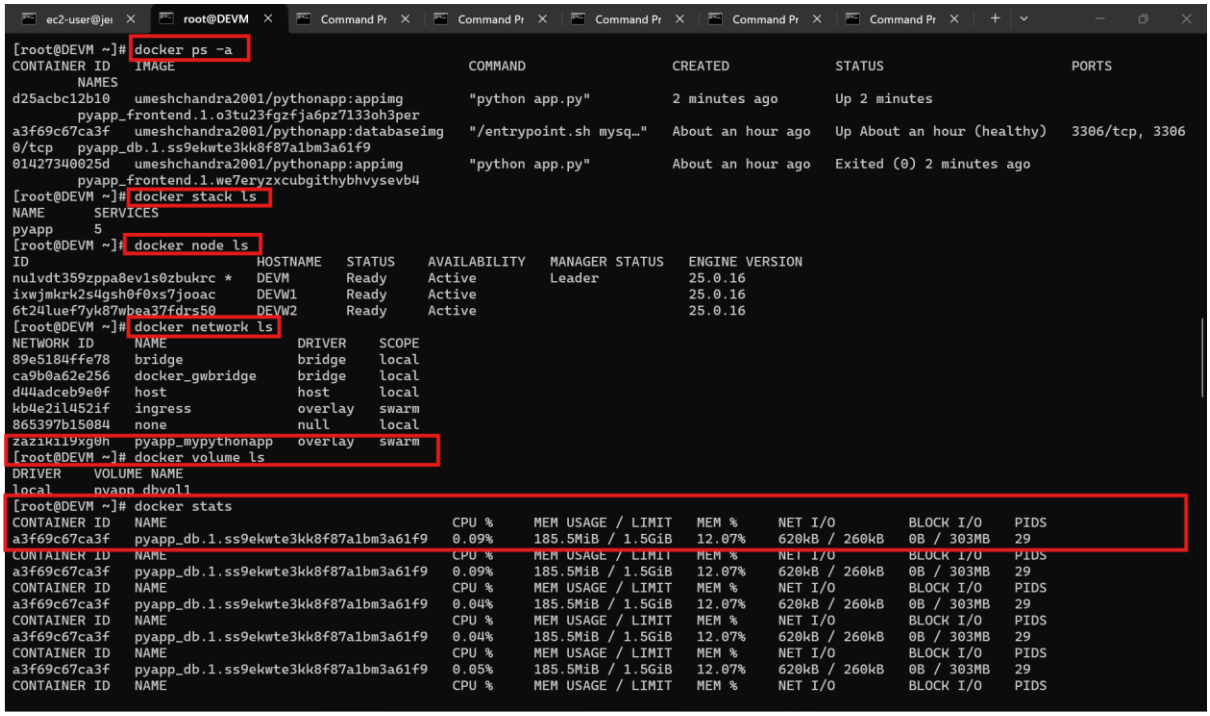
- Triggers
    - Build after other projects are built
        - Give the previous build node name
        - Trigger only if previous build is stable (Success)

Dockerhub Repo:
http://hub.docker.com/repository/docker/umeshchandra2001/pythonapp

APPLICATION WORKING VIDEO:
https://drive.google.com/file/d/1m43hTzntY6GnLqF0Lr72oqmtbxJOZIeQ/view?usp=drive_link

Pipeline Automation video URL:
https://drive.google.com/file/d/1PwSkZIfl0o3jJi81bPAULHURL-B-rRcM/view?usp=sharing

UAT IMAGES:



PRD SERVER IMAGES

